

## **MSML606 Extra Credit Project 2 Proposal**

### **Team Composition:**

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### **AI Usage Statement:**

I used AI to understand which type of algorithm is suitable for different kinds of problems. I used Kaggle to find the suitable dataset for my project. I have manually texted the document.

### **Dataset:**

This project will use the Amazon 10,000 products Dataset, which contains metadata for 10,005 real world consumer products.

Each record has attributes such as product name, category, price, weight, and average customer rating. This dataset contains more than 10,000 records and includes multiple metadata fields beyond the primary key. This dataset comes from real world Amazon listings, it contains noisy and inconsistent formatting, including unparsed weight strings, missing price values, and special characters in the product titles. This makes the dataset suitable for evaluating how algorithms perform on real-world logistics data rather than clean, academic datasets.

### **Dataset Source:**

Amazon 10,000 Products Dataset:

<https://www.kaggle.com/datasets/nguyenngocphung/10000-amazon-products-dataset>

### **Application Description:**

The objective of this project is to build a Logistics Load Optimizer using the 0/1 Knapsack Algorithm. This algorithm uses a Dynamic programming approach to efficiently search the products for a shipping container.

In this system, every container has a fixed weight capacity to store the product. Each product from the dataset will be evaluated for inclusion in a container with a fixed weight capacity. The product's price acts as its "Value," and its extracted weight acts as its "Weight". When a user inputs a specific container weight limit, the system will execute the Knapsack algorithm and return the optimal Shipping Manifest.

For example, if a user sets a capacity of 50,000g, the system will return:

Total Items Packed: 52

Total Shipment Value: \$3,450

Capacity Utilization: 98.5%

As this project uses Space Optimized Dynamic Programming, the operations will run efficiently even with over 10,000 records. By utilizing a 1D array instead of a 2D matrix, the algorithm will reduce the space complexity from  $O(N \times W)$  to  $O(W)$  where  $N$  is the number of items and  $W$  is the weight of items, making it more efficient for large-scale logistics than a simple linear or brute-force approach.

Before running the optimization, the dataset has to undergo various preprocessing steps as it is real world data. These include converting price strings into integers, using regular expressions to extract the numerical weights from the product descriptions, handling missing values, and normalizing product titles. These steps ensure that the dataset can be processed by the DP engine without any errors.

Finally, the project will evaluate the efficiency of the Space-Optimized DP system by comparing its performance and cargo value against a Baseline Greedy Algorithm.